

Experiment No.9 Letter Pattern in Seven Segment Display Application																																																																																																				
Course Code: CPE 013	Program: BSCpE																																																																																																			
Course Title: Logic Circuits and Design	Date Performed: 17/3/2026																																																																																																			
Section: BSCpE22S2	Date Submitted: 19/3/2026																																																																																																			
Student Name: Bulaong, Ross Andrew D. Florencio, David Malabanan, Nadine Que, Leroy Josef Zapanta, Russell	Instructor: Engr. Noel Lusung																																																																																																			
1. Objective(s): This activity aims to design combinational logic circuits using logic gates, decoder and K-Mapping to desired display text in seven-segment.																																																																																																				
2. Intended Learning Outcomes (ILOs): The students shall be able to: 2.1 Use K-Map in generating the expected output generated in TinkerCAD application. 2.2 Apply decoder combinational logic circuit with the Seven Segment display.																																																																																																				
3. Discussion: <i>Observe and analyze the given problem and generating its solution:</i> SAMPLE PROBLEM WITH SOLUTION: Problem: Design a 3 – bit input logic circuit that will display JULY 30 in seven segment. SOLUTION: <table border="1" style="margin-top: 10px; width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr> <th></th> <th style="color: blue;">A</th> <th style="color: blue;">B</th> <th style="color: blue;">C</th> <th style="color: blue;">a</th> <th style="color: blue;">b</th> <th style="color: blue;">c</th> <th style="color: blue;">d</th> <th style="color: blue;">e</th> <th style="color: blue;">f</th> <th style="color: blue;">g</th> </tr> </thead> <tbody> <tr> <td style="color: blue;">0</td> <td>0</td> <td>0</td> <td>0</td> <td style="color: red;">0</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">0</td> <td style="color: red;">0</td> </tr> <tr> <td style="color: blue;">1</td> <td>0</td> <td>0</td> <td>1</td> <td style="color: red;">0</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">0</td> </tr> <tr> <td style="color: blue;">2</td> <td>0</td> <td>1</td> <td>0</td> <td style="color: red;">0</td> <td style="color: red;">0</td> <td style="color: red;">0</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">0</td> </tr> <tr> <td style="color: blue;">3</td> <td>0</td> <td>1</td> <td>1</td> <td style="color: red;">0</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">0</td> <td style="color: red;">1</td> <td style="color: red;">1</td> </tr> <tr> <td style="color: blue;">4</td> <td>1</td> <td>0</td> <td>0</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">0</td> <td style="color: red;">0</td> <td style="color: red;">1</td> </tr> <tr> <td style="color: blue;">5</td> <td>1</td> <td>0</td> <td>1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">1</td> <td style="color: red;">0</td> </tr> <tr> <td style="color: blue;">6</td> <td>1</td> <td>1</td> <td>0</td> <td colspan="7" style="background-color: black;"></td> </tr> <tr> <td style="color: blue;">7</td> <td>1</td> <td>1</td> <td>1</td> <td colspan="7" style="background-color: black;"></td> </tr> </tbody> </table>			A	B	C	a	b	c	d	e	f	g	0	0	0	0	0	1	1	1	1	0	0	1	0	0	1	0	1	1	1	1	1	0	2	0	1	0	0	0	0	1	1	1	0	3	0	1	1	0	1	1	1	0	1	1	4	1	0	0	1	1	1	1	0	0	1	5	1	0	1	1	1	1	1	1	1	0	6	1	1	0								7	1	1	1							
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7	1	1	1																																																																																																	

a	BC	00	01	11	10
A					
0					
1			1	1	d

$$a(ABC) = \sum \{4, 5\}$$

$$a = L_1$$

$$= A$$

b	BC	00	01	11	10
A					
0			1	1	1
1			1	1	d

$$b(ABC) = \sum \{0, 1, 3, 4, 5\}$$

$$b = L_1 + L_2$$

$$= \bar{B} + C$$

c	BC	00	01	11	10
A					
0			1	1	1
1			1	1	d

$$c(ABC) = \sum \{0, 1, 3, 4, 5\}$$

$$c = L_1 + L_2$$

$$= \bar{B} + C$$

d	BC	00	01	11	10
A					
0			1	1	1
1			1	1	d

$$d(ABC) = \sum \{0, 1, 2, 3, 4, 5\}$$

$$d = VCC$$

e	BC	00	01	11	10
A					
0			1	1	1
1				1	d

$$e(ABC) = \sum \{0, 1, 2, 5\}$$

$$e = L_1 + L_2 + L_3$$

$$= \bar{A}\bar{B} + \bar{B}C + \bar{A}\bar{C}$$

$$= \bar{A}(\bar{B} + \bar{C}) + \bar{B}C$$

f	BC	00	01	11	10
A					
0			1	1	1
1			1	d	d

$$f(ABC) = \sum \{1, 2, 3, 5\}$$

$$f = L_1 + L_2$$

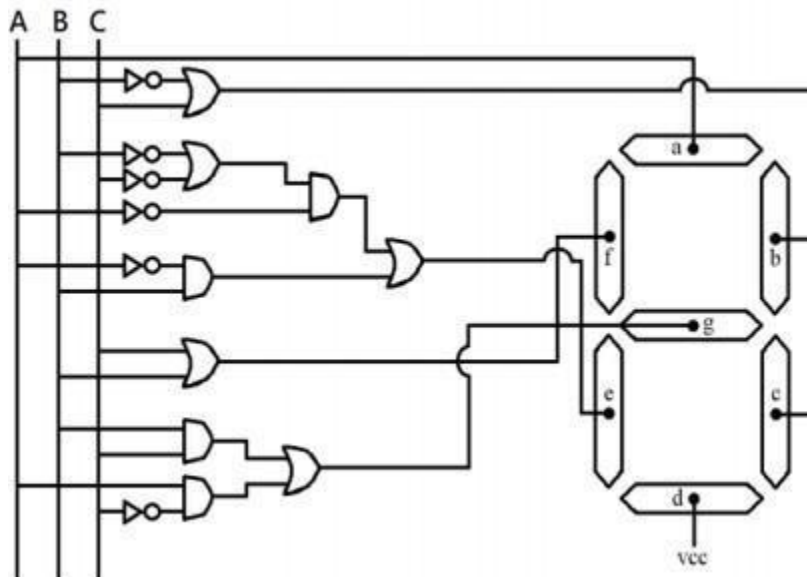
$$= B + C$$

g	BC	00	01	11	10
A					
0				1	
1		1		d	d

$$g(ABC) = \sum \{3, 4\}$$

$$g = L_1 + L_2$$

$$= A\bar{C} + BC$$



4. Resources:

- Personal Computer
 - Internet connection
- Simulation Tools:
- o TinkerCAD
 - o Logicly

5. Procedure:

1. Analyze the given problem and determine the requirements (number of inputs and number of outputs) of the problem.
2. Construct the truth table.
3. Simplify the output function using K-Mapping. Show this as your solution.
4. Draw the circuit diagram.
5. Construct the circuit and simulate its operation in Logic.ly by verifying the input / output relationship.
6. Simulate the operation in TinkerCAD by connecting the inputs, wires, electronic components, and output correctly following the pin configuration of the ICs.
7. After the pasting an output screenshot and writing down the link for the uploaded video recording below, write the conclusion, and answer the questions on section 8.

Problem No. 1

Design a combinational logic circuit that will display CPE013L in a seven segment following the inputs

(A,B,C) below. Show solution using K-Map then draw Logic circuit diagram. See image below for your letter reference in 7-Segment.

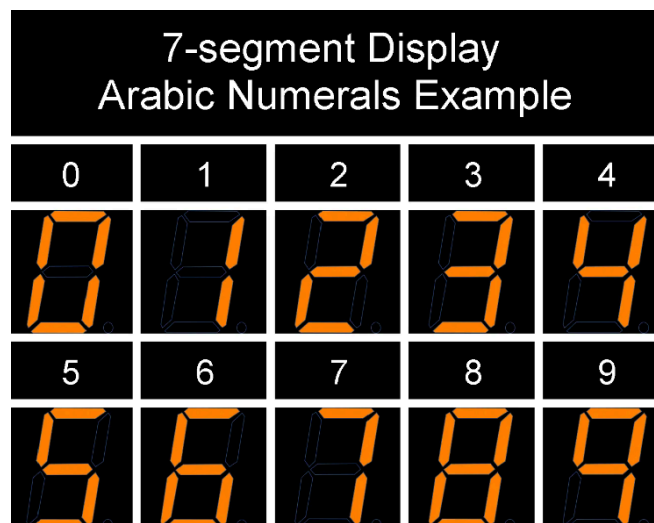
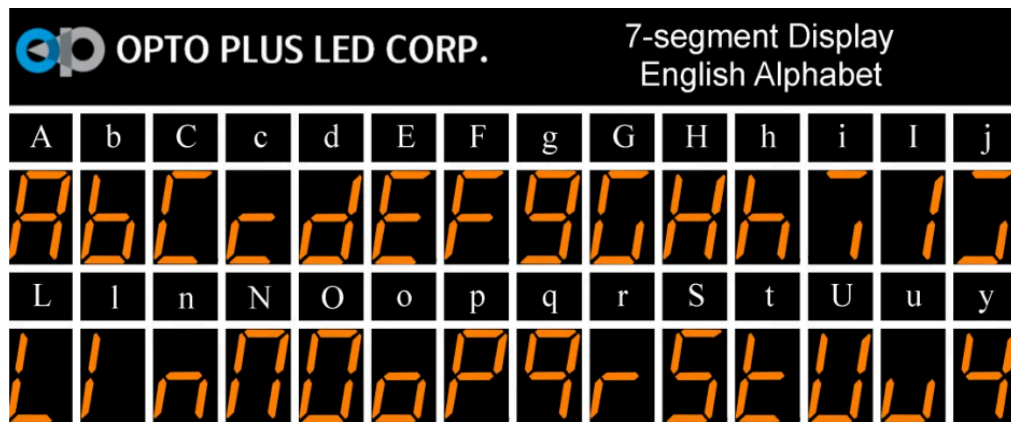


Image Source: <https://www.opledtw.com/blog/7-segment-16-segment-represent-numbers-letters-explained/>

KMAP

a					
	A/BC	00	01	11	10
	0	1	1	1	1
	1	0	1		

Simplified Boolean Expression = $A' + B'C$

b					
	A/BC	00	01	11	10
	0	0	1	1	0
	1	1	1		

Simplified Boolean Expression = $AB' + B'C + A'C$

c					
	A/BC	00	01	11	10
	0	0	0	1	0
	1	1	1		

Simplified Boolean Expression = $AB' + A'BC$

d					
	A/BC	00	01	11	10
	0	1	0	1	1
	1	0	1		

Simplified Boolean Expression = $A'B'C' + AB'C + A'B$



e					
	A/BC	00	01	11	10
	0	1	1	1	1
	1	0	0		

Simplified Boolean Expression = A'

f					
	A/BC	00	01	11	10
	0	1	1	1	1
	1	0	0		

Simplified Boolean Expression = A'

g	  				
	A/BC	00	01	11	10
	0	0	1	0	1
	1	0	1		

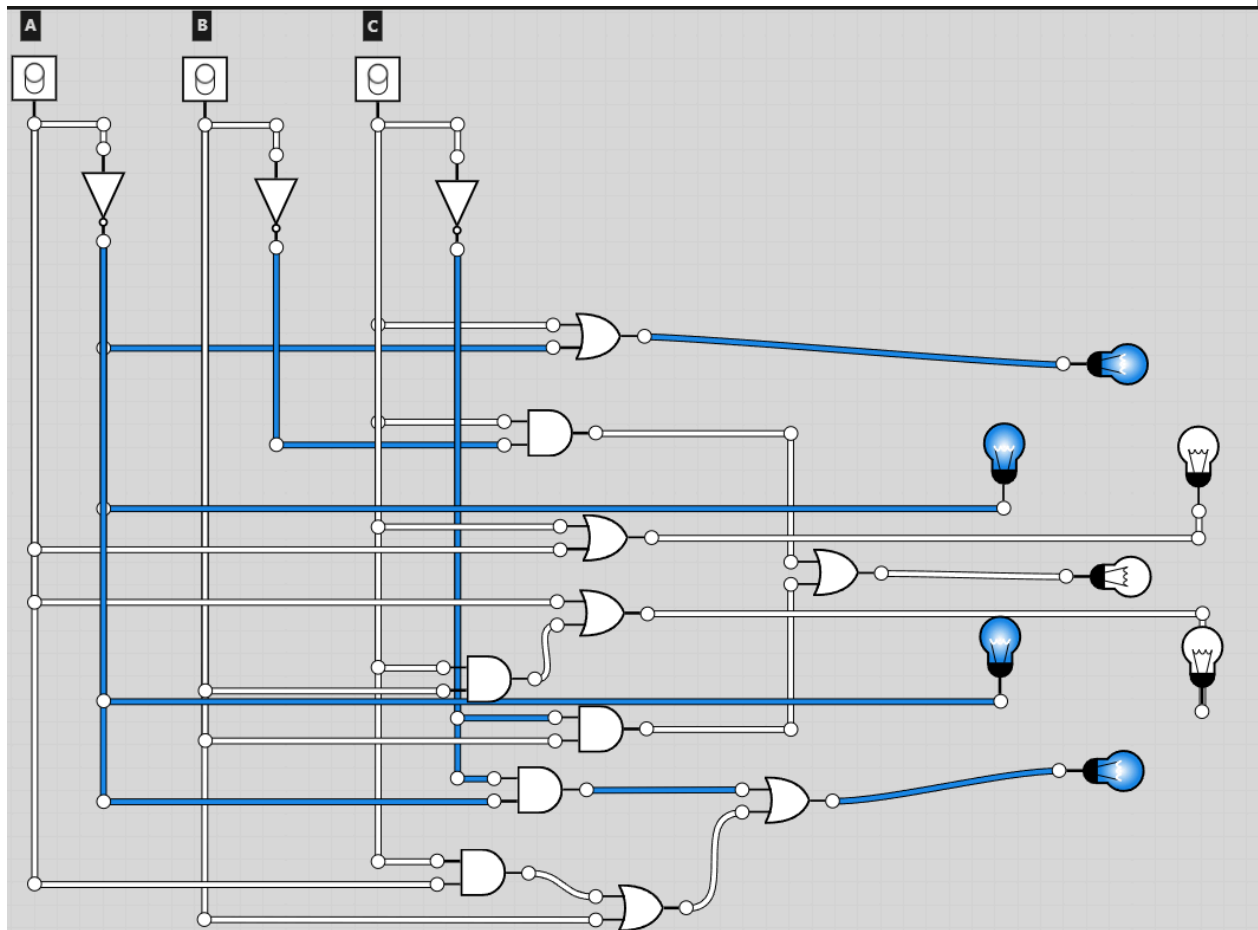


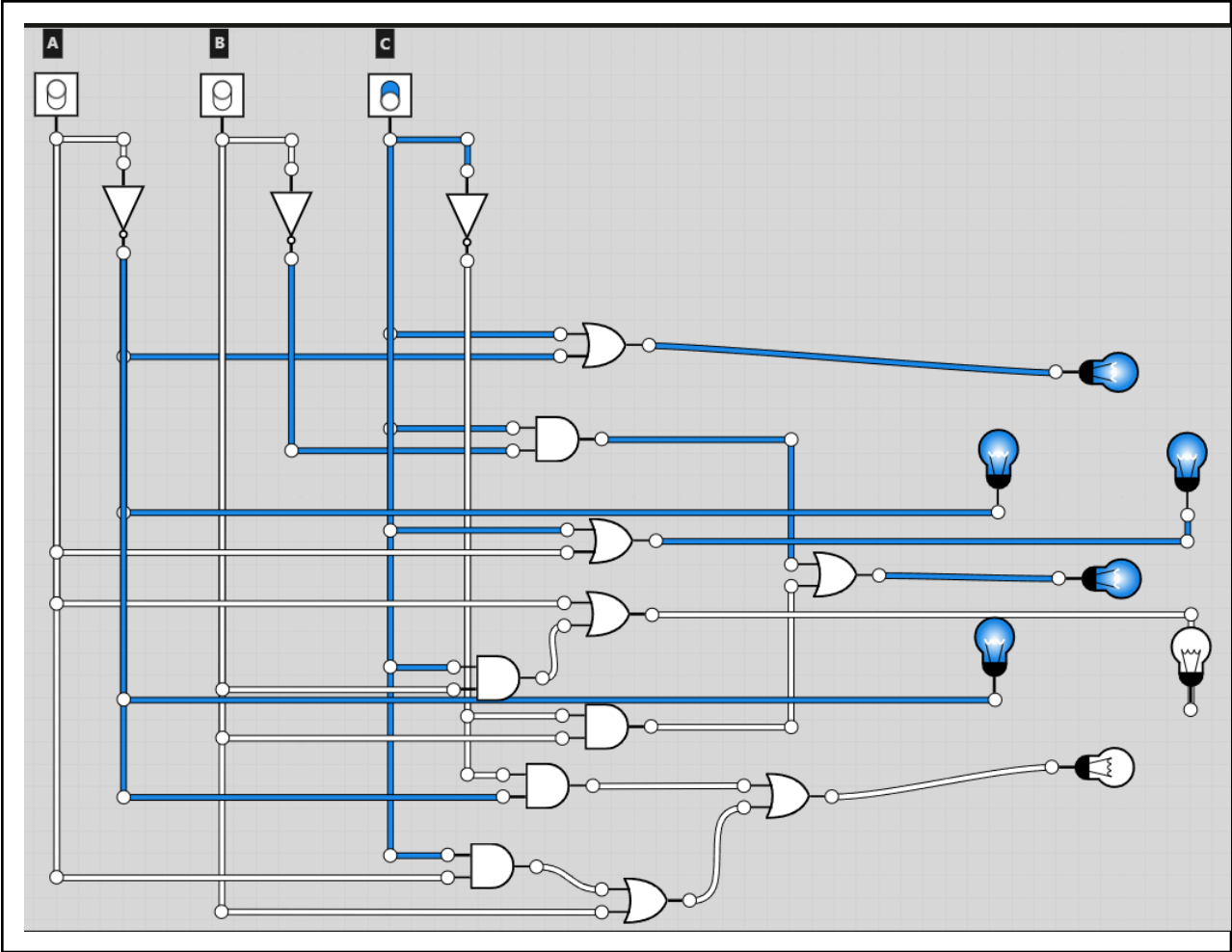
Simplified Boolean Expression = $B'C + A'BC'$

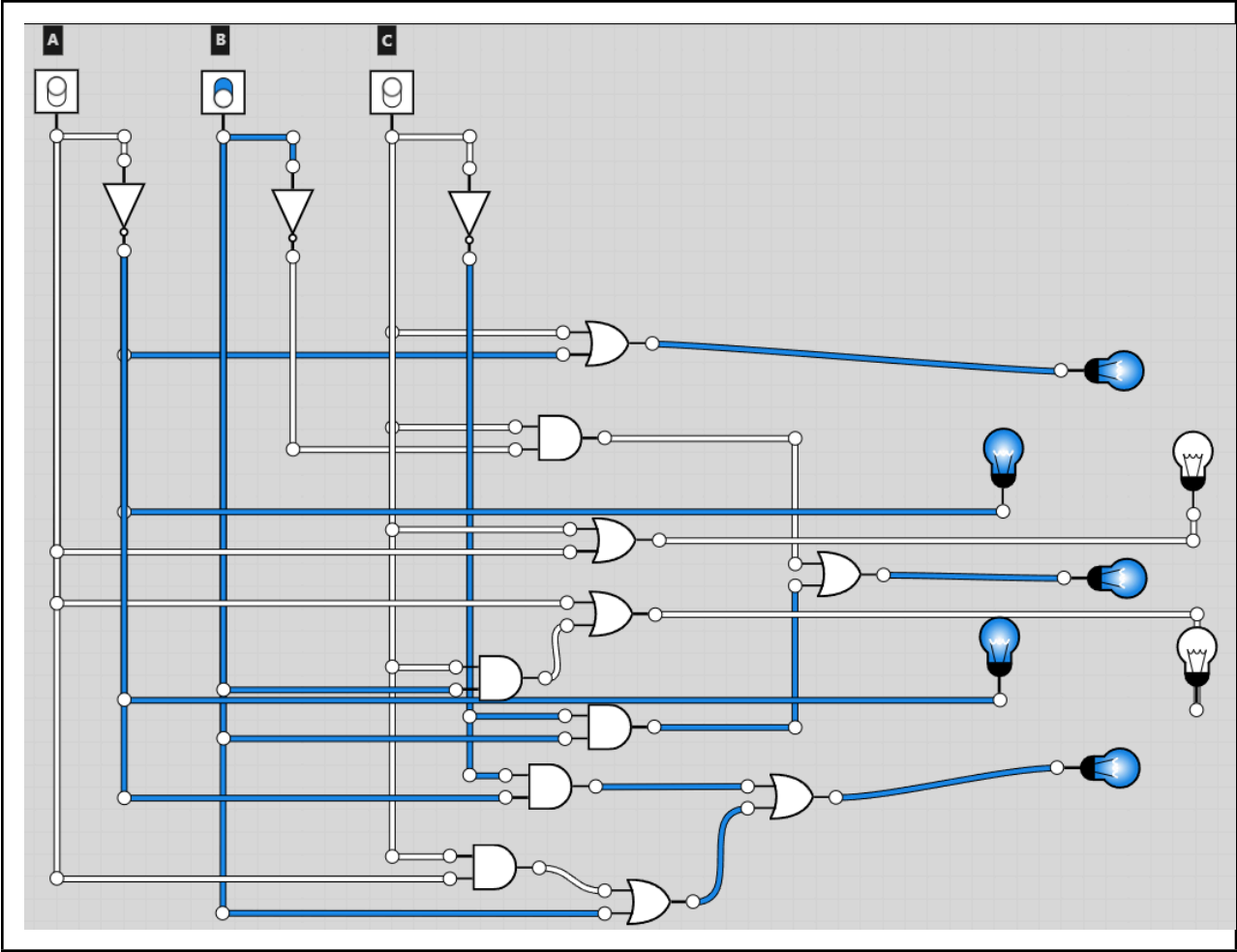
Truth Table

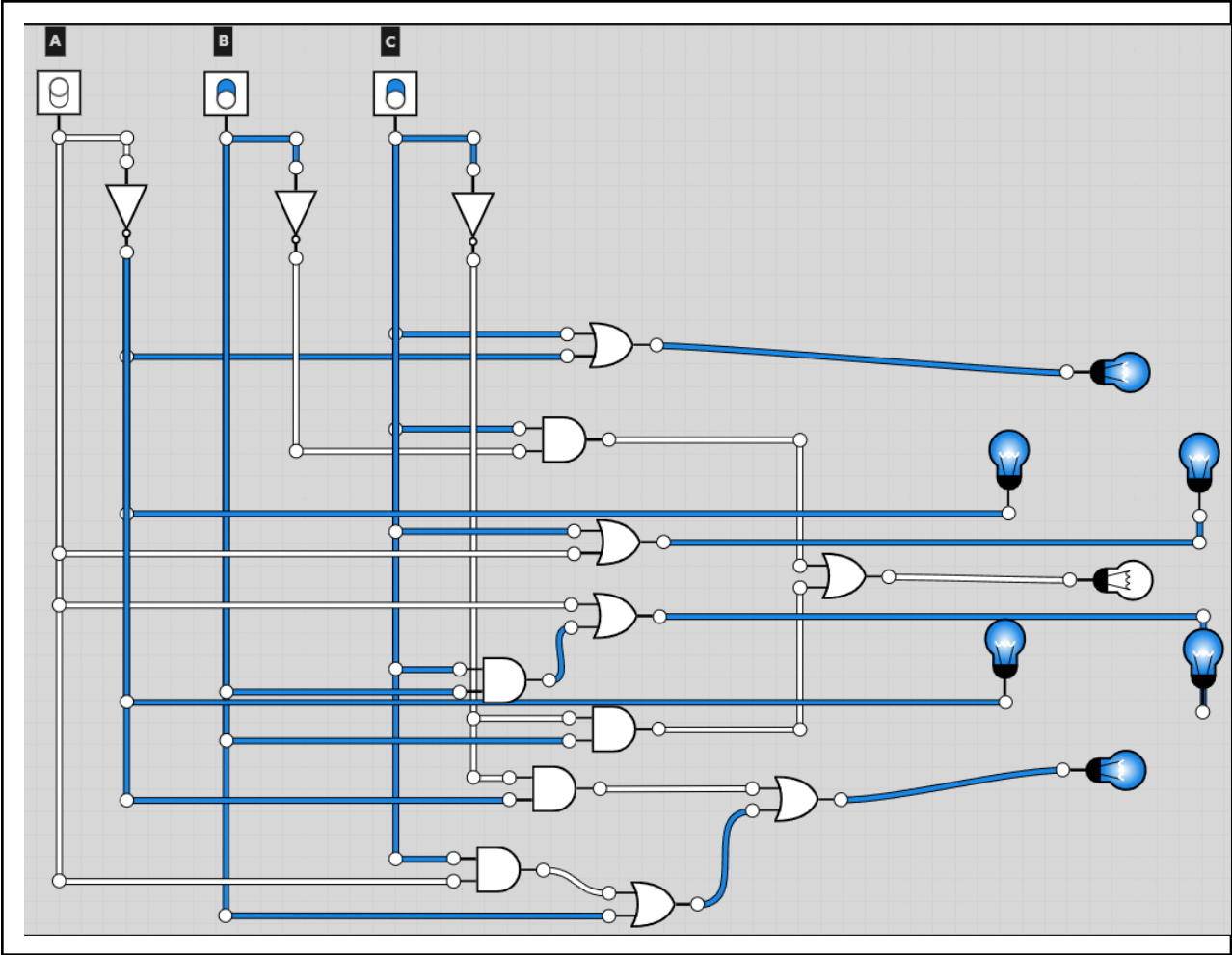
A	B	C	a	b	c	d	e	f	g
0	0	0	1	0	0	1	1	1	0
0	0	1	1	1	0	0	1	1	1
0	1	0	1	0	0	1	1	1	1
0	1	1	1	1	1	1	1	1	0
1	0	0	0	1	1	0	0	0	0
1	0	1	1	1	1	1	0	0	1

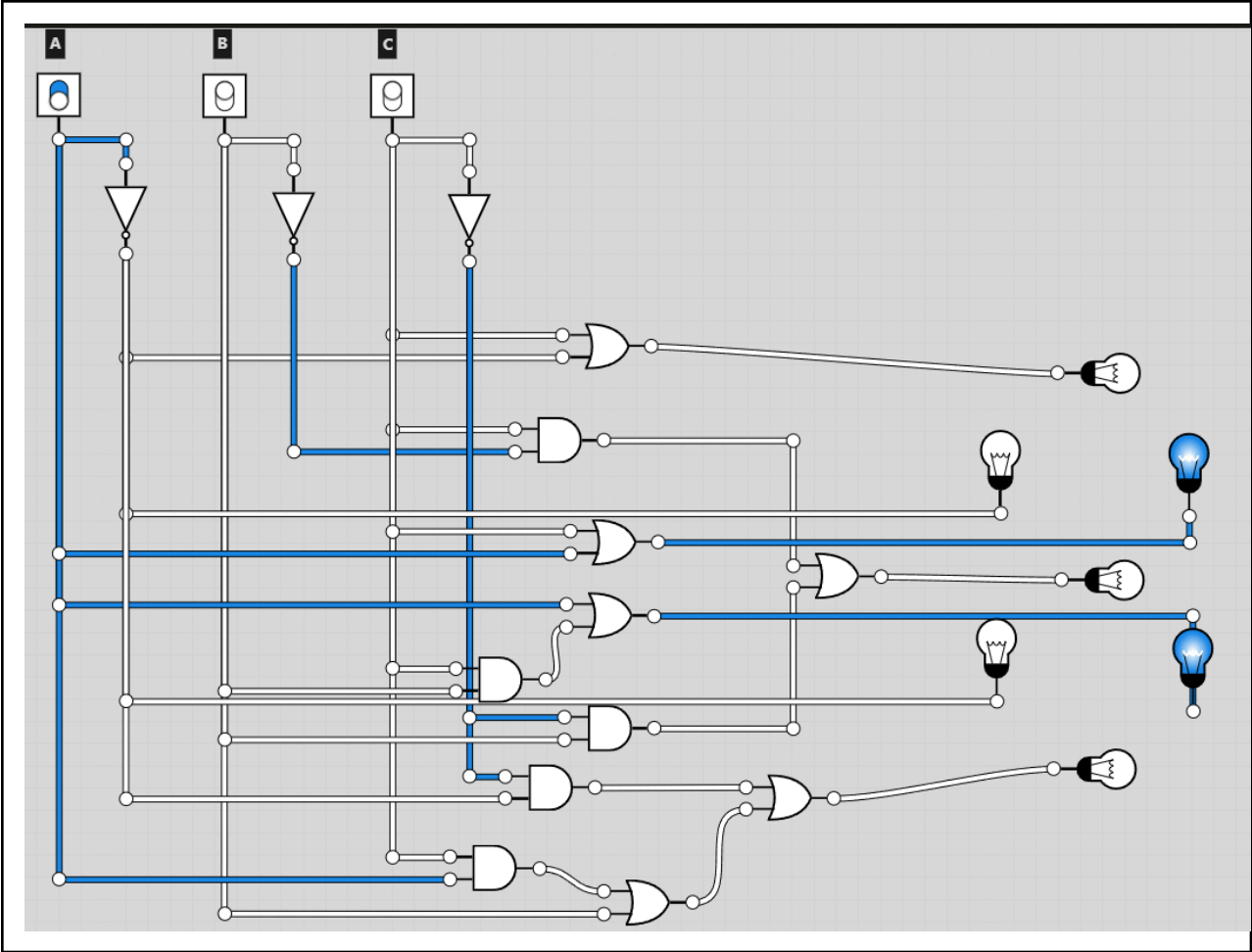
Logicly for CPE013:

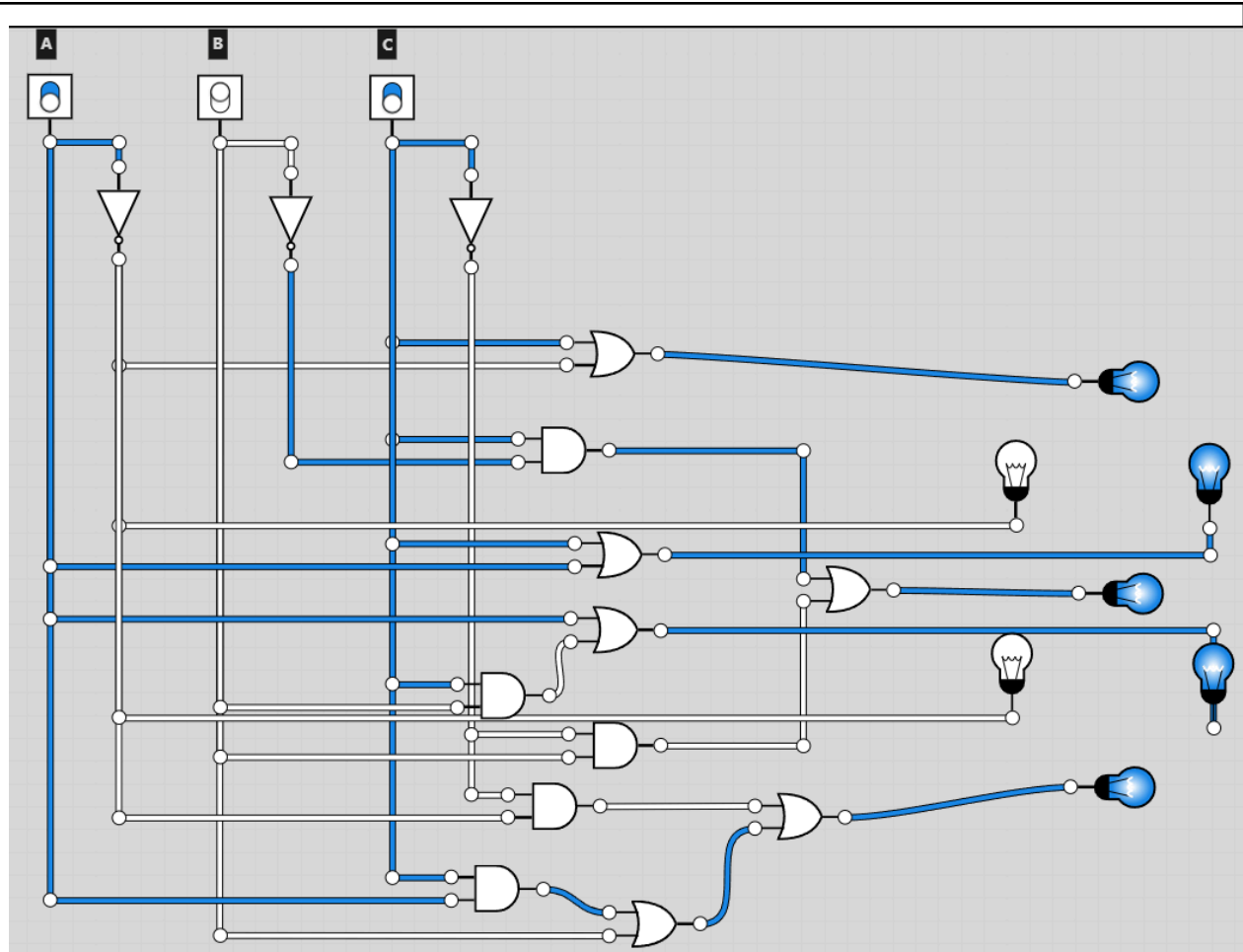




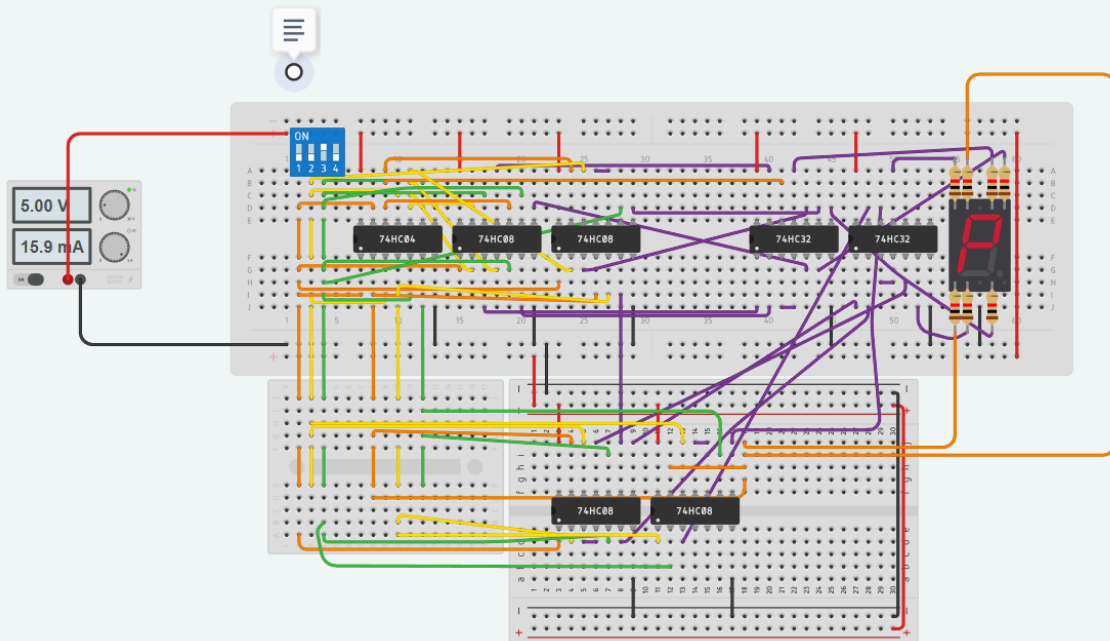
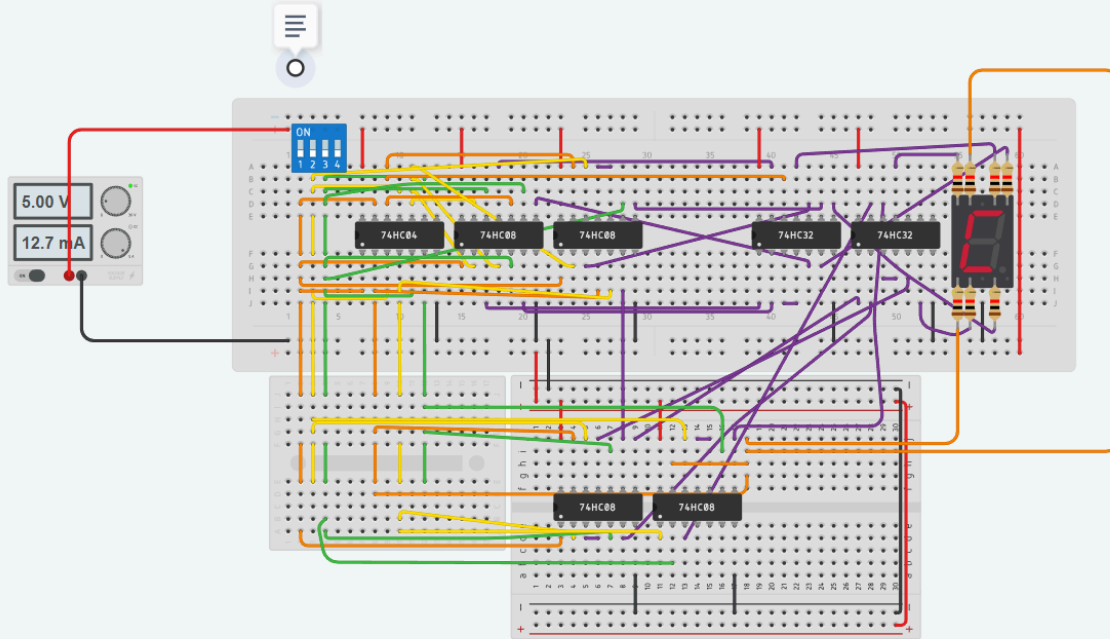


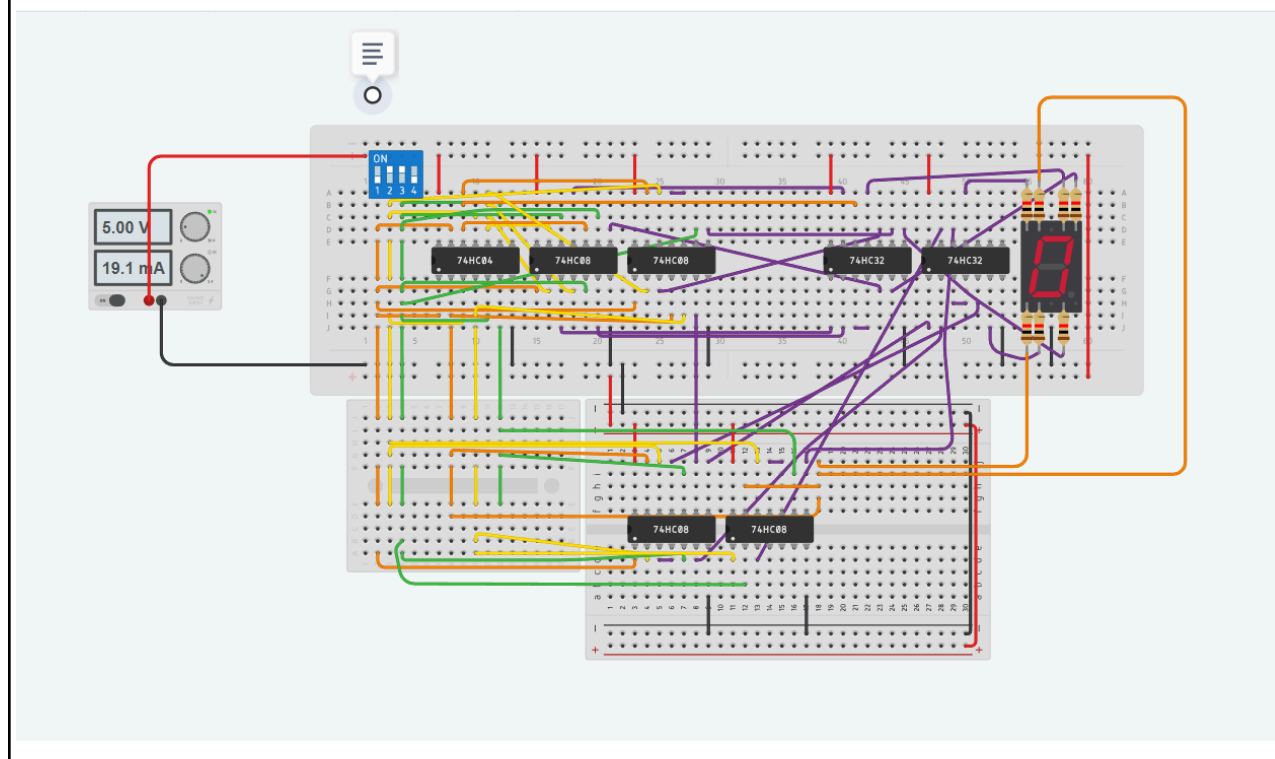
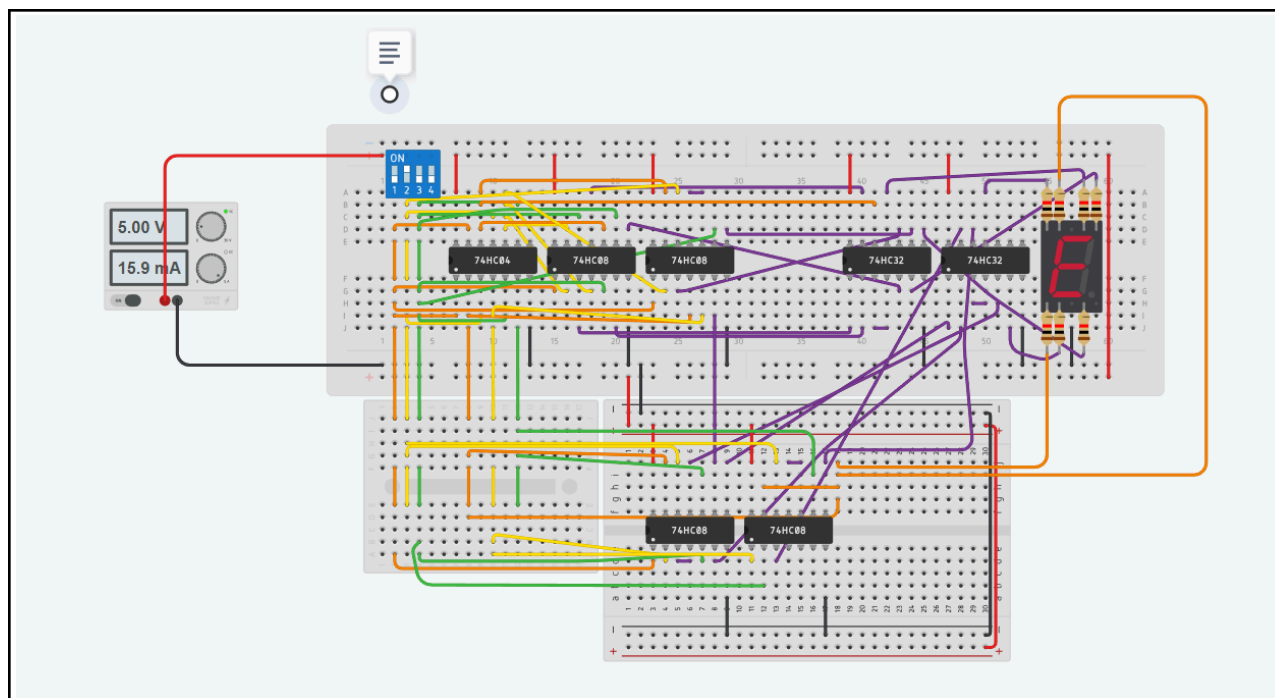


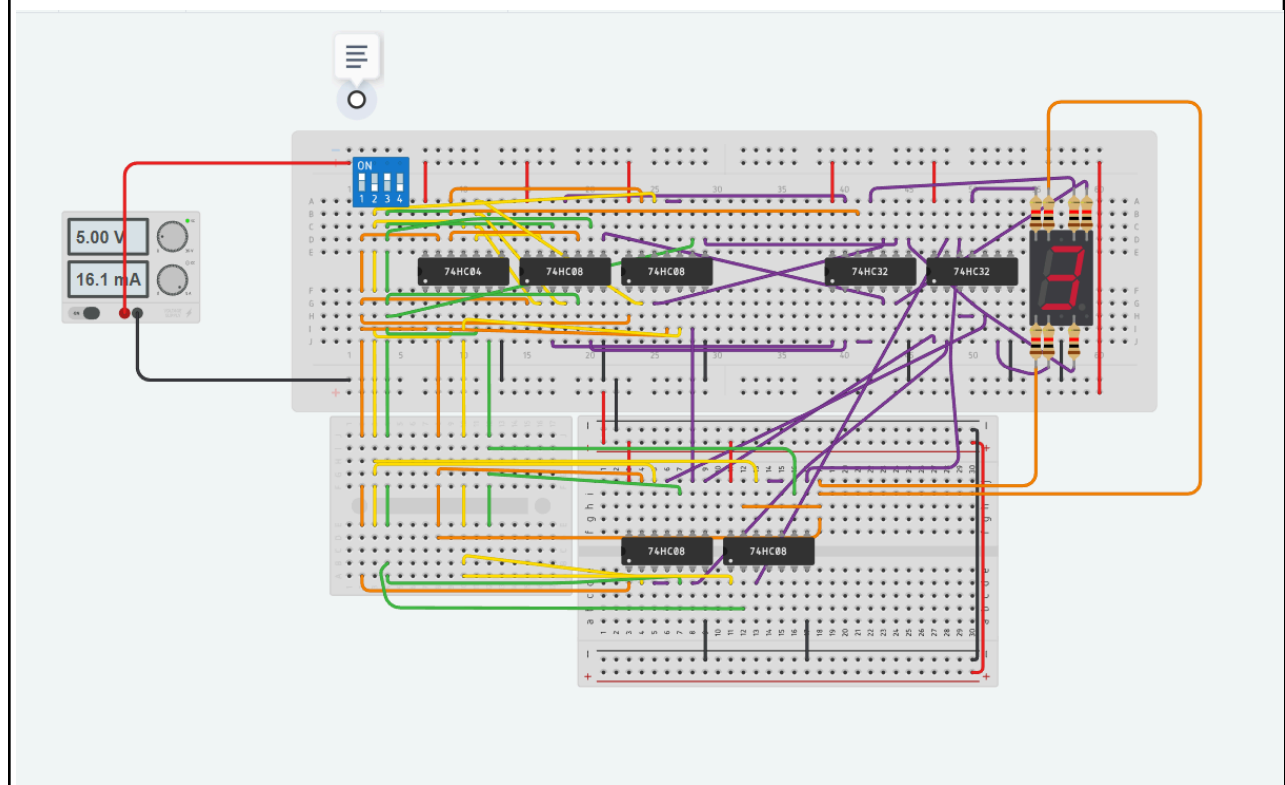
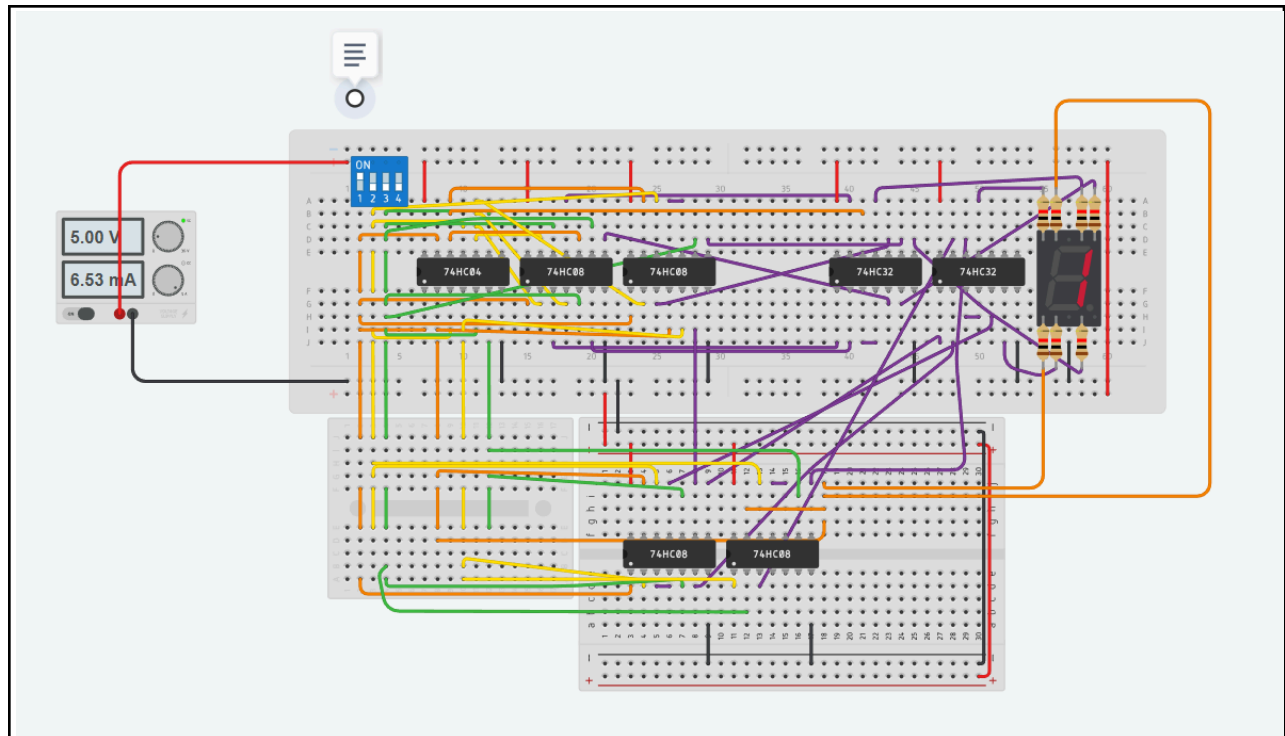




TinkerCAD for CPE013:







6. Final Output:

Screenshot or print screen of TinkercAD application different output simulations (5 points deductions for only one output in every required circuit). Also, upload a video of the simulated circuit naming or talking about the shown input combinations. Failure to comply to each instruction will result to 5 points

deductions. Please include the recorded simulation video in this section and write at least one sentence definition or label for each figure or image.

TINKERCAD CIRCUIT:

<https://www.tinkercad.com/things/bLDkpzFNAYt-cpe013-letter-pattern-in-7-segment-display>

LOGICLY and TINKERCAD VIDEO DEMO:  experiment9vids

7. Conclusion

Write a one paragraph (at least 10 sentences) Conclusion for this activity.

In this experiment, we learned how to design a combinational logic circuit using logic gates, decoders, and K-Map to display characters on a seven-segment display. The problem was analyzed to determine the required inputs and outputs. A truth table was constructed to understand the relationship between input combinations and the output segments. Using K-Map, the Boolean expressions were simplified to create a more efficient circuit. The simplified expressions were then translated into a logic circuit diagram. The circuit was tested using Logic.ly to verify that it produces the correct outputs. TinkerCAD was also used to properly connect components based on the correct pin configuration. Some difficulties were encountered during the process, but they were resolved through careful analysis and adjustments. This activity improved understanding of combinational circuits and logic design. Overall, the activity successfully demonstrated the application of theoretical concepts in practical circuit simulation.

8. Questions:

1. What is the primarily function of a Decoder circuit?

The primary function of a decoder circuit is to convert a binary input code into the unique set of segment activation signals (a through g) required to display the desired character on a seven-segment display.

The decoder takes a binary input (A, B, C, etc.) which represents the desired digit or character. It then activates the specific segment outputs (a, b, c, d, e, f, g) required to display that digit or character on the seven-segment display. It essentially translates the compact binary input into the specific segment signals needed for the display.

2. What is the purpose of a, b, c, d, e, f, and g in seven segment display?

The letters a, b, c, d, e, f, and g represent the seven individual Light Emitting Diodes (LEDs), or segments, that make up the display. Their purpose is to illuminate in different combinations to display decimal numerals (0-9) and, in some cases, letters or other characters.

9. Assessment (Rubric for Laboratory Performance):

Laboratory Performance Rubric (2)					
Criteria	Ratings				Pts
Manipulative Skills	3 pts Proficient Student always demonstrate needed skills.	2 pts Acceptable Student occasionally demonstrate needed skills.	1 pts Beginner Student do not demonstrate needed skills.		3 pts
Experimental Set - up	3 pts Proficient Student are able to set-up the material with minimum supervision.	2 pts Acceptable Student are able to set-up the materials with supervision.	1 pts Beginner Student are unable to set-up the materials.		3 pts
Process Skills	3 pts Proficient Student always demonstrate targeted process skills.	2 pts Acceptable Student occasionally demonstrate targeted process skills.	1 pts Beginner Student do not demonstrate targeted process skills.		3 pts
Application of Technical Writing Skills	3 pts Proficient Correct spelling, grammar, and sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	2 pts Acceptable Some minor incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	1 pts Beginner Many incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.		3 pts
Time Management / Conduct of Experiment	3 pts Proficient Student finish ahead of time with complete data and time to revise data.	2 pts Acceptable Student finish on time with incomplete data.	1 pts Beginner Student do not finish on time with incomplete data.		3 pts
Complete Requirements of Laboratory Report	3 pts Proficient Student submitted a complete requirements set by the teacher.	2 pts Acceptable Student submitted an almost complete Laboratory Report according to requirements set by the teacher.	1 pts Beginner Student submitted a lot of missing requirements in the Laboratory Report according to requirements set by the teacher.		3 pts
Follow the Correct Format of Laboratory Report	3 pts Proficient Student submitted a correct format of Laboratory Report according to requirements set by the teacher.	2 pts Acceptable Student submitted an almost correct format of Laboratory Report according to requirements set by the teacher.	1 pts Beginner Student did not follow at all the format of Laboratory Report according to requirements set by the teacher.		3 pts
Ability to do independent work	3 pts Proficient Student do not need to be supervised by the teacher.	2 pts Acceptable Student require occasional supervision by the teacher.	1 pts Beginner Student require supervision by the teacher.		3 pts
Total Points: 24					